

Docker Cheat Sheet

Basics

```
docker --version      # Check Docker version
docker info           # Show system-wide information
docker help           # Show help for Docker commands
```

Images

```
docker pull nginx      # Download an image from Docker Hub
docker images          # List all local images
docker rmi nginx       # Remove an image
docker rmi -f <image_id> # Force remove an image
```

Containers

```
docker ps             # List running containers
docker ps -a          # List all containers (including stopped)
docker run hello-world # Run a test container
docker run -it ubuntu bash # Run Ubuntu interactively
docker run -d -p 8080:80 nginx # Run in detached mode with port mapping
docker stop <container_id> # Stop a container
docker start <container_id> # Start a stopped container
docker restart <container_id> # Restart a container
docker rm <container_id> # Remove a stopped container
docker rm -f <container_id> # Force remove a container
```

Logs & Exec

```
docker logs <container_id> # Show container logs
docker logs -f <container_id> # Follow container logs
docker exec -it <container_id> bash # Access a running container
```

Volumes

```
docker volume create mydata # Create a volume
docker volume ls            # List volumes
docker volume rm mydata     # Remove a volume
docker volume prune         # Remove unused volumes
```

Networks

```
docker network ls        # List networks
docker network create mynet # Create a network
docker network connect mynet <container_id> # Connect container to network
docker network rm mynet   # Remove a network
```

System Cleanup

```
docker system df         # Show disk usage
docker system prune      # Remove unused data
docker system prune -a --volumes # Aggressive cleanup (everything unused)
```

Build

```
docker build -t myapp:1.0 . # Build image from Dockerfile
docker tag myapp:1.0 myrepo/myapp:1.0 # Tag an image
docker push myrepo/myapp:1.0 # Push image to Docker Hub
```

Docker Compose

```
docker-compose up          # Start services from docker-compose.yml
docker-compose up -d       # Detached mode
docker-compose down        # Stop and remove containers/networks
docker-compose logs -f     # Show logs for all services
```

Advanced

```
docker stats               # Show live resource usage
docker top <container_id>  # Show processes inside container
docker inspect <container_id> # Detailed container info (JSON)
docker cp file.txt <container_id>:/app # Copy file into container
docker commit <container_id> myapp:2.0 # Save container as new image
```